

Quiz 1

BSSE23-A

Multivariable Calculus | (MT101T) | Fall '25

Name: _____ Reg. No.: _____

Q1: [CLO-1] [3 pts]

Find r so that the vector from the point $A (1, -1, 3)$ to the point $B (3, 0, 5)$ is orthogonal to the vector from A to the point $P(r, r, r)$.

Q2: [CLO-1] [4 pts]

Find the orthogonal projection of $\mathbf{v} = \mathbf{i} + \mathbf{j} + \mathbf{k}$ on $\mathbf{b} = 2\mathbf{i} + 2\mathbf{j}$, and then find the vector component of $\mathbf{v}$ orthogonal to $\mathbf{b}$.

Q3: [CLO-1] [3 pts]

Where does the line $x = 1 + t, y = 3 - t, z = 2t$ intersect the cylinder $x^2 + y^2 = 16$?